Required Capstone Checkpoint 1.1: Evaluating When Retrieval Is Required

Directions

Select one of the provided capstone scenarios and use it as the foundation for this checkpoint. You will continue developing this same scenario throughout the program. Download this worksheet and save a copy to your device before you begin working. Type your responses directly into the document, save your completed work, and upload the finished file when you are ready to submit.

Use this worksheet to:

- Analyze how an LLM performs without retrieval
- Document prompts and corresponding model outputs
- Identify limitations and patterns in model behavior
- Determine when retrieval is required
- Reflect on how retrieval quality affects reliability and user trust

1. Selected Scenario and Intended Use

Select your capstone scenario:

☐ Research Paper Navigator

☒ **Wikipedia Retrieval Engine**

Briefly describe the kinds of questions, tasks, or user interactions your system should support.
The system should support factual, comparative, explanatory, and multi-document questions about significant people, places, and topics. Users should also be able to ask follow-up questions and request source quotations to verify answers. Answers must be grounded.

MIGUEL HERRERA – SECTION B – August 26, 2026 – Checkpoint 1.1

2. Prompt and Output Evidence

- Provide two prompts relevant to your scenario. Where possible, use prompts that test different types of information needs or tasks.
- Include the corresponding LLM outputs for each prompt.

Prompt 1 and Corresponding LLM Outputs **SCENARIO=wikipedia MODEL=openai/gpt-5.4-mini**
PROMPT 4/9: Who was Albert Einstein?... **RESPONSE**: Albert Einstein was a theoretical physicist best known for developing the theory of relativity, including the famous equation \\(E = mc^2\\), which shows the relationship between mass and energy. He was born in 1879 in Ulm, in what is now Germany, and later became one of the most influential scientists in history. His work helped reshape modern physics, especially our understanding of space, time, gravity, and light. Einstein also won the 1921 Nobel Prize in Physics, not for relativity, but for his explanation of the photoelectric effect.

Prompt 2 and Corresponding LLM Outputs (Follow-up question) **SCENARIO=wikipedia MODEL=openai/gpt-5.4-mini**
PROMPT 5/9: What he was especially notable for?... **RESPONSE**: Who are you referring to? If you share the person's name or the document/context, I can answer accurately.

Prompt 3 and Corresponding LLM Outputs (Follow-up question) SCENARIO=wikipedia MODEL=openai/gpt-5.4-mini
PROMPT 6/9: Did he receive any gifts from his father when young?... **RESPONSE**: I'm not sure who "he" refers to or which document/story you mean. If you share the relevant passage or tell me the person's name, I can answer directly and quote the text.

3. Analysis of Model Limitations

- Identify key limitations in the model's responses, such as hallucinations, knowledge gaps, stale information, inconsistency, vagueness, or missing information.
- Use specific prompt/output examples to support your analysis.

Key limitations and supporting prompt/output examples
The LLM model shows the following limitations: **Missing Information Example** **PROMPT 9/9**: How many weapons are featured in the video game 'GoldenEye 007'?... **RESPONSE**: I'm not certain from the information I have here. If you mean the Nintendo 64 video game **GoldenEye 007**, it features a large arsenal, but I don't have a reliable source in this context to give an exact weapon count without guessing. If you want, I can help you find the exact number from a specific source or list the weapons included in the game. **Knowledge Gaps Example** **PROMPT 8/9**: How is Ace Bailey related to it?... **RESPONSE**: I'm not sure what "it" refers to here. If you mean a specific document, event, team, or person, please point me to it and I can explain how Ace Bailey is related.

4. Patterns in Model Behavior

- Describe any meaningful patterns or trends you observe across the outputs.
- Consider whether certain types of prompts produce more reliable responses than others.

Meaningful patterns or trends
Response: • **Pattern**: The model loses the context of a series of questions: When I ask a follow up question that includes *pronouns* such as "he" or "it" the model loses the "context" of the conversation and does not provide an answer. • **Trend**: The model tends to ask for additional documents or clarification when it cannot provide a meaningful/complete answer.

5. Determining Whether Retrieval Is Required

- Explain whether retrieval is required for your scenario and identify which observed limitations retrieval could help address.
- Justify your reasoning using evidence from the outputs.

Explanation and justification
Retrieval-augmented generation (RAG) would be needed in this scenario because the LLM has difficulty answering questions when information is missing from its context and when follow-up questions depend on previously retrieved information. The outputs shows that the model may acknowledge when it lacks reliable information and request additional sources, while also struggling to resolve pronouns such as "he" or "it" in follow-up questions. RAG can address these limitations by retrieving relevant information from a **trusted knowledge source** and providing that context to the LLM before generating a response, enabling more accurate, grounded answers and better continuity across related questions. Evidence Output SCENARIO=wikipedia MODEL=openai/gpt-5.4-mini == PROMPT 4/9: Who was Albert Einstein?... RESPONSE: Albert Einstein was a theoretical physicist best known for developing the theory of relativity, including the famous equation \(E = mc^2\), which shows the relationship between mass and energy. He was born in 1879 in Ulm, in what is now Germany, and later became one of the most influential scientists in history. His work helped reshape modern physics, especially our understanding of space, time, gravity, and light. Einstein also won the 1921 Nobel Prize in Physics, not for relativity, but for his explanation of the photoelectric effect. == == PROMPT 5/9: What he was especially notable for?...

RESPONSE:

Who are you referring to? If you share the person’s name or the document/context, I can answer accurately.

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PROMPT 6/9: Did he receive any gifts from his father when young?...

RESPONSE:

I’m not sure who “he” refers to or which document/story you mean.

If you share the relevant passage or tell me the person’s name, I can answer directly and quote the text.

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PROMPT 7/9: What is Toronto Maple Leafs...

RESPONSE:

The **Toronto Maple Leafs** are a professional **ice hockey team** based in **Toronto, Ontario, Canada**. They play in the **National Hockey League (NHL)** and are one of the league’s “Original Six” teams.

If you want, I can also give you:

- a short history of the team,

- their current roster,

- or explain why they’re called the Maple Leafs.

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PROMPT 8/9: How is Ace Bailey related to it?...

RESPONSE:

I’m not sure what “it” refers to here. If you mean a specific document, event, team, or person, please point me to it and I can explain how Ace Bailey is related.

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PROMPT 9/9: How many weapons are featured in the video game 'GoldenEye 007'?...

RESPONSE:

I’m not certain from the information I have here. If you mean the Nintendo 64 video game **GoldenEye 007**, it features a large arsenal, but I don’t have a reliable source in this context to give an exact weapon count without guessing.

If you want, I can help you find the exact number from a specific source or list the weapons included in the game.

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6. Retrieval Requirements

- Describe the types of information that would need to be retrieved to improve reliability, grounding, and response quality. Consider whether the information is proprietary, frequently changing, highly specific, or otherwise unavailable to the model without retrieval.
- Explain when retrieval would or would not be necessary.

Information that would need to be retrieved
Response: The system should retrieve <u>relevant facts</u>, <u>authoritative sources</u> (i.e. Trusted reference source), and rely on conversational context (remembering its history) to fill knowledge gaps, resolve references, and provide grounded, reliable answers.

When retrieval would or would not be necessary
Retrieval is needed when the model cannot: • Provide relevant facts. • Quote sources or information that is accurate and reliable. • Follow-up questions Retrieval would not be necessary for simple questions when the LLM already has <u>sufficient context</u> and reliable knowledge to answer accurately.

7. Reflection and Next Steps

Explain how the precision, relevance, and accuracy of retrieved information affect output quality, system reliability, and user trust. Consider how retrieval quality might improve or degrade system performance. What could happen if the retrieved information is incomplete, loosely related, outdated, or inaccurate?

Explanation
Retrieval quality directly affects system reliability and user trust. For example, when information is missing, such as the exact number of weapons in *GoldenEye 007*, the model's credibility can be reduced. Therefore, poor retrieval, such as incomplete, irrelevant, outdated, or inaccurate information can produce unreliable answers and reduce user trust. RAG can help fill these knowledge gaps by providing accurate, relevant, and complete information, allowing the LLM to generate more grounded and dependable responses.

8. Retrieval Requirements

- Reflect on what you learned about LLM limitations, common failure modes, and retrieval needs from this activity.
- Describe what you would investigate or improve next as you continue developing your capstone system.

Reflection
Response: I learned that some key LLM limitations include: - **Hallucinations**: Incorrect or fabricated facts that sound confident. - **Staleness**: Generating answers from outdated information. - **Provide inaccurate answers** from "guessing" the next word instead of basing its answer from real world facts. I learned that Retrieval is needed when the model cannot provide relevant facts, or generate answers that are "relevant" to the user's input.

Supporting Materials

Use this section to list, describe, or link to any supporting materials included with your submission.

Supporting material	Location, link, or note
CAPSTONE: GITHUB REPOSITORY	https://github.com/miguelangelherrera77-blip/MIT_CAPSTONE/tree/main/Final_Capstone_Project/Capstone_Checkpoint_1.1
Google Gemini answer for: *"How many weapons are featured in the video game GoldenEye 007"*	https://share.google/aimode/vl8PduWi6UrCVX7I4 "The original 1997 Nintendo 64 video game **GoldenEye 007** features **32 weapons**, including unique items like Unarmed and the Detonator…"

End of Required Capstone Checkpoint 1.1

Return to the assignment page in the learning platform to upload and submit your completed capstone checkpoint and complete any remaining items in this module.